JEE Main Online Exam 2020

Questions & Solution

9th January 2020 | Shift - I

PHYSICS

Q.1 If the screw on a screw-gauge is given six rotations, it moves by 3 mm on the main scale. If there are 50 divisions on the circular scale the least count of the screw gauge is-

(1) 0.001 cm (2) 0.02 mm (3) 0.01 cm (4) 0.001 mm

Ans. **[1]**

Sol. $\text{Pitch} = \frac{3}{6}\ \text{mm} = 0.5\ \text{mm}$

$\text{Least count} = \frac{0.5}{50}\ \text{mm}$

$= 0.001\ \text{cm}$

Q.2 Radiation, with wavelength 6561 Å falls on a metal surface to produce photoelectrons. The electrons are made to enter a uniform magnetic field of 3×10^{-4} T. If the radius of the largest circular path followed by the electrons is 10 mm, the work function of the metal is close to-

(1) 0.8 eV (2) 1.6 eV (3) 1.1 eV (4) 1.8 eV

Ans. **[3]**

Sol. $h\nu - \phi = kE_{max.}$; $h\nu = \frac{12400}{6561} = 1.9\ \text{eV}$

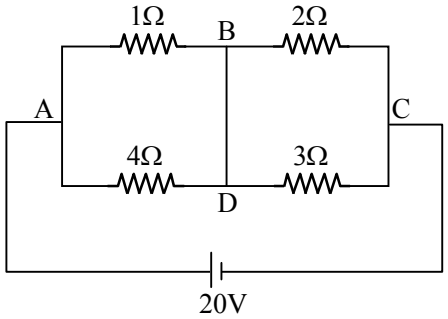
$r = \frac{mv}{qB}$; $p = mv = rqB$

$kE_{max.} = \frac{p^2}{2m} = \frac{q^2r^2B^2}{2m}\text{J} = \frac{(1.6\times10^{-19})^2(10\times10^{-3})^2\times(3\times10^{-4})^2}{2\times9\times10^{-31}\times1.6\times10^{-19}}\ \text{eV}$

$\approx 0.8\ \text{eV}$

$\phi = h\nu - kE_{max.} = 1.9 - 0.8 = 1.1\ \text{eV}$

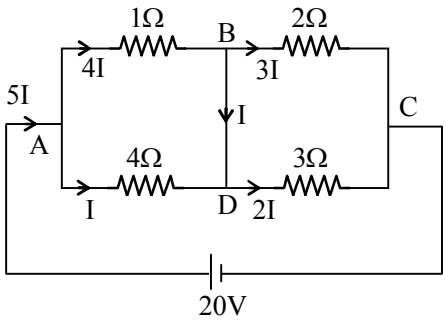
Q.3 In the given circuit diagram, a wire is joining points B and D. The current in this wire is-



(1) 4 A (2) 0.4 A (3) 2 A (4) zero

Ans. **[3]**

Sol.



$20 - 4I - 6I = 0 \Rightarrow \boxed{I = 2A}$

Q.4 Two particals of equal mass m have respective initial velocities $u\hat{i}$ and $u\left(\frac{\hat{i}+\hat{j}}{2}\right)$. They collide completely inelastically. The energy lost in the process is-

(1) $\frac{3}{4}mu^2$ (2) $\frac{1}{8}mu^2$ (3) $\frac{1}{3}mu^2$ (4) $\sqrt{\frac{2}{3}}mu^2$

Ans. **[2]**

Sol. $m u\hat{i} + m\left(\frac{u}{2}\hat{i} + \frac{u}{2}\hat{j}\right) = 2m\vec{v}$

$\vec{v} = \frac{3u}{4}\hat{i} + \frac{u}{4}\hat{j}$

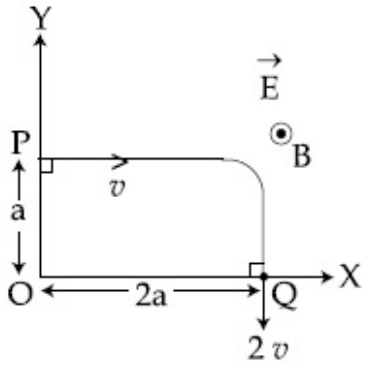
$|\vec{v}|^2 = \frac{10u^2}{16}$

$\Delta KE = KE_f - KE_i$

$= \frac{1}{2}2m \times \frac{10u^2}{16} - \frac{1}{2}mu^2 - \frac{1}{2}m\frac{u^2}{2}$

$\Delta KE = \frac{-mu^2}{8}$

Q.5 A charged particle of mass 'm' and charge 'q' moving under the influence of uniform electric field $E\vec{i}$ and a uniform magnetic field $B\vec{k}$ follows a trajectory from point P to Q as shown in figure. The velocities at P and Q are respectively, $v\vec{i}$ and $-2v\vec{j}$. Then which of the following statements (A, B, C, D) are the correct ? (Trajectory shown in schematic and not to scale)



(A) $E = \frac{3}{4}\left(\frac{mv^2}{qa}\right)$

(B) Rate of work done by the electric field at P is $\frac{3}{4}\left(\frac{mv^3}{a}\right)$

(C) Rate of work done by both the fields at Q is zero

(D) The difference between the magnitude of angular momentum of the particle at P and Q is 2 mav.

(1) B, C, D (2) A, B, C (3) A, B, C, D (4) A, C, D

Ans. **[2]**

Sol. Work done by magnetic force = zero

Thus

ΔKE = Work done by electric field

$$\frac{1}{2}m(2v)^2 - \frac{1}{2}mv^2 = qE \times 2a$$

$$\Rightarrow \boxed{E = \frac{3}{4}\frac{mv^2}{qa}}$$

Rate of work done by electric field at p = qEv = $\frac{3}{4}\frac{mv^3}{a}$

Rate of work done by magnetic field at any point is zero since force $\perp$ velocity, at Q $\vec{E}$ is perpendicular to velocity thus rate of work done by it is zero.

Q.6 A body A of mass m is moving in a circular orbit of radius R about a planet. Another body B of mass $\frac{m}{2}$ collides with A with a velocity which is half $\left(\frac{\vec{v}}{2}\right)$ the instantaneous velocity $\vec{v}$ of A. The collision is completely inelastic. Then, the combined body-

(1) starts moving in an elliptical orbit around the planet

(2) continues to move in a circular orbit

(3) falls vertically downwards towards the planet

(4) escapes from the planet's gravitational field

Ans. **[1]**

Sol. Orbital speed for orbital of A is v

After collision

$$mv + \frac{m}{2} \times \frac{v}{2} = \frac{3m}{2}v' \; ; \; v' = \frac{5}{6}v$$

$\Rightarrow v' < v \Rightarrow$ path will be elliptical with center of earth at the near focus.

Q.7 A quantity f is given by $f = \sqrt{\frac{hc^5}{G}}$ where c is speed of light, G universal gravitational constant of light and h is the Planck's constant. Dimension of f is that of-

(1) area (2) volume (3) momentum (4) energy

Ans. **[4]**

Sol. $f = \sqrt{\frac{hc^2}{G}}$

$[h] = [M^1L^2T^{-1}]$

$[c] = [L^1T^{-1}]$

$[G] = [M^{-1}L^3T^{-2}]$

$[f] = \sqrt{\frac{[M^1L^2T^{-1}][L^5T^{-5}]}{[M^{-1}L^3T^{-2}]}} = [M^1L^2T^{-2}]$

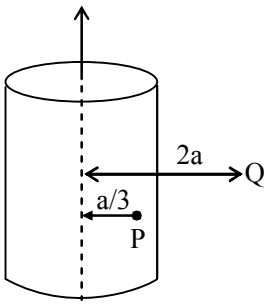
[f] is equal to dimension of energy.

Q.8 A long, straight wire of radius a carries a current distributed uniformly over its cross-section. The ratio of the magnetic fields due to wire at distance $\frac{a}{3}$ and 2a, respectively from the axis of the wire is-

(1) $\frac{1}{2}$ (2) 2 (3) $\frac{2}{3}$ (4) $\frac{3}{2}$

Ans. **[3]**

Sol.



Let current density be J.

For point P

$\oint \vec{B}.d\vec{\ell} = \mu_0 i$

$B_P 2\pi\frac{a}{3} = \mu_0 J\pi\frac{a^2}{9}$

For point Q

$B_Q 2\pi(2a) = \mu_0 J\pi a^2$

$\frac{B_P\ 2\pi\frac{a}{3}}{B_Q\ 4\pi a} = \frac{\mu_0 J\pi\frac{a^2}{9}}{\mu_0 J\pi a^2}$

$\Rightarrow \frac{B_P}{B_Q} = \frac{2}{3}$

Q.9 An electric dipole of moment $\vec{p} = (-\hat{i} - 3\hat{j} + 2\hat{k}) \times 10^{-29}$ C.m is at the origin (0, 0, 0). The electric field due to this dipole at $\vec{r} = +\hat{i} + 3\hat{j} + 5\hat{k}$ (note that $\vec{r}.\vec{p} = 0$ is parallel to-

(1) $(+\hat{i} + 3\hat{j} - 2\hat{k})$ (2) $(+\hat{i} - 3\hat{j} - 2\hat{k})$ (3) $(-\hat{i} - 3\hat{j} + 2\hat{k})$ (4) $(-\hat{i} + 3\hat{j} - 2\hat{k})$

Ans. **[1]**

Sol. Since $\vec{r}.\vec{p} = 0$ $\Rightarrow$ it is equatorial position

$\Rightarrow \vec{E}$ will be antiparallel to $\vec{p}$

Q.10 A particle moving with kinetic energy E has de Broglie wavelength λ. If energy ΔE is added to its energy, the wavelength become λ/2. Value of ΔE, is-

(1) 3E (2) E (3) 4E (4) 2E

Ans. **[1]**

Sol. $\lambda = \dfrac{h}{\sqrt{2mE}}$

$\lambda' = \dfrac{h}{\sqrt{2mE'}}$

$\lambda' = \dfrac{\lambda}{2} \Rightarrow E' = 4E$

$\Delta E = 4E - E = 3E$

Q.11 The aperture diameter of a telescope is 5 m. The separation between the moon and the earth is 4×10^5 km. With light of wavelength of 5500 Å, the minimum separation between objects on the surface of moon, so that they are just resolved, is close to-

(1) 20 m (2) 200 m (3) 60 m (4) 600 m

Ans. **[3]**

Sol. $\theta = 1.22 \dfrac{\lambda}{a}$

$d_{min} = \theta D$

$= \dfrac{1.22 \lambda D}{a}$

$\lambda = 5500 \times 10^{-10}$

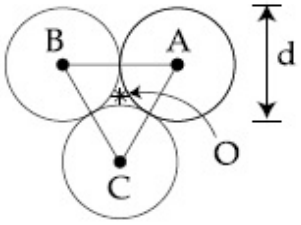
$D = 4 \times 10^8$ m

$a = 5m$

$d_{min} = \dfrac{1.22 \times 5500 \times 10^{-10} \times 4 \times 10^8}{5} = 53.6$ m

Correct answer = 60 m

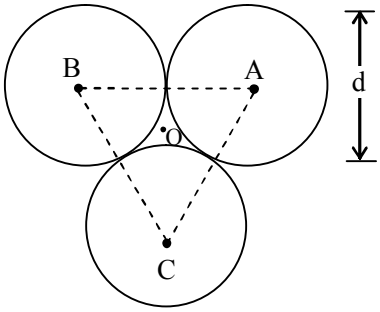
Q.12 Three solid spheres each of mass m and diameter d are struck together such that the lines connecting the centres form an equilateral triangle of side of length d. The ratio I_0/I_A of moment of inertia I_0 of the system about an axis passing the centroid and about center of any of the spheres I_A and perpendicular to the plane of the triangle is-



(1) $\frac{13}{15}$ (2) $\frac{15}{13}$ (3) $\frac{23}{13}$ (4) $\frac{13}{23}$

Ans. **[4]**

Sol.



Distance ; AB = d

$$OA = \frac{d}{\sqrt{3}}$$

$$I_0 = 3\left[\frac{2}{5}m\left(\frac{d}{2}\right)^2 + m\left(\frac{d}{\sqrt{3}}\right)^2\right] = \frac{13\,md^2}{10}$$

$$I_A = \frac{2m}{5}\left(\frac{d}{2}\right)^2 + 2\left[\frac{2}{5}m\left(\frac{d}{2}\right)^2 + m(d)^2\right] = \frac{23md^2}{10}$$

$$\boxed{\frac{I_0}{I_A} = \frac{13}{23}}$$

Q.13 Consider two identical diatomic gases A and B at some temperature T. Molecules of the gas A are rigid, and have a mass m. Molecules of the gas B have an additional vibrational mode, and have a mass $\frac{m}{4}$. The ratio of the specific heats (C_V^A and C_V^B) of gas A and B, respectively is-

(1) 5 : 9 (2) 7 : 9 (3) 3 : 5 (4) 5 : 7

Ans. **[4]**

Sol. $f_A = 5$ $f_B = 7$

$$C_V^A = \frac{5R}{2} \qquad C_V^B = \frac{7R}{2}$$

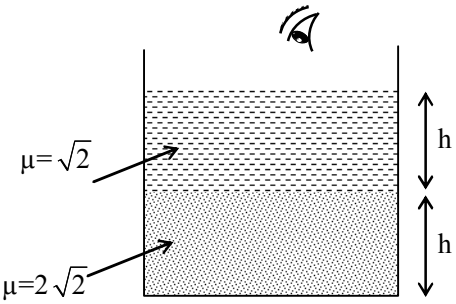
$$\boxed{\text{Ratio} = \frac{5}{7}}$$

Q.14 A vessel of depth 2h is half filled with a liquid of refractive index $2\sqrt{2}$ and the upper half with another liquid of refractive index $\sqrt{2}$. The liquids are immiscible. The apparent depth of the inner surface of the bottom of vessel will be-

(1) $\frac{3}{4}h\sqrt{2}$ (2) $\frac{h}{2(\sqrt{2}+1)}$ (3) $\frac{h}{3\sqrt{2}}$ (4) $\frac{h}{\sqrt{2}}$

Ans. **[1]**

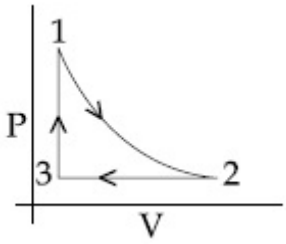
Sol.



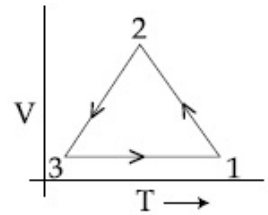
$$d = \frac{h}{2\sqrt{2}} + \frac{h}{\sqrt{2}} = \frac{3h}{2\sqrt{2}}$$

$$d = \frac{3\sqrt{2}\,h}{4}$$

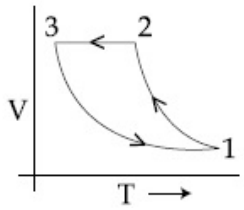
Q.15 Which of the following is an equivalent cyclic process corresponding to the thermodynamics cyclic given in the figure ? Where, $1 \to 2$ is adiabatic. (Graphs are schematic and are not to scale)



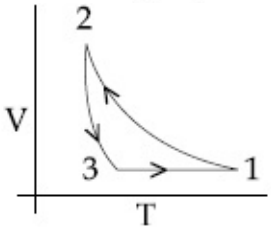
(1)



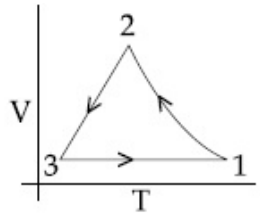
(2)



(3)



(4)



Ans. **[4]**

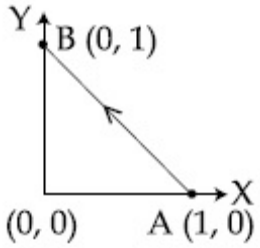
For process $1 \to 2$ $TV^{\gamma-1} = \text{const.}$

For process $2 \to 3$ $\frac{V}{T} = \text{const.}$

For process $3 \to 1$ $V = \text{const.}$

Thus correct ans. is option 4

Q.16 Consider a force $\vec{F} = -x\hat{i} + y\hat{j}$. The work done by this force in moving a particle from point A(1, 0) to B(0, 1) along the line segment is (all quantities are in SI units)



(1) 2 (2) 1 (3) $\frac{1}{2}$ (4) $\frac{3}{2}$

Ans. **[2]**

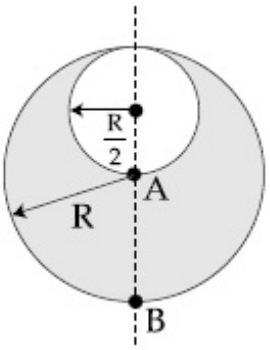
$\vec{F} = -x\hat{i} + y\hat{j}$

$W = \int \vec{F} \cdot d\vec{r} = \int_1^0 -x\,dx + \int_0^1 y\,dy$

$= \frac{1}{2} + \frac{1}{2}$

W = 1 J

Q.17 Consider a sphere of radius R which carries a uniform charge density ρ. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\frac{|\vec{E}_A|}{|\vec{E}_B|}$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is-



(1) $\frac{17}{54}$ (2) $\frac{18}{54}$ (3) $\frac{18}{34}$ (4) $\frac{21}{34}$

Ans. **[3]**

$E_A = \frac{\rho R/2}{3\varepsilon_0} = \frac{\rho R}{6\varepsilon_0}$

$$E_B = \frac{E\left(\rho\frac{4}{3}\pi R^3\right)}{R^2} - \frac{k\left(\rho\frac{4}{3}\pi\frac{R^3}{8}\right)}{\left(\frac{3R}{2}\right)^2} = \frac{17\rho R}{54\varepsilon_0}$$

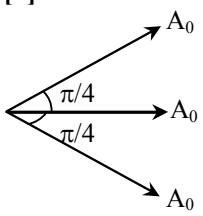
$$\frac{E_A}{E_B} = \frac{\rho R/6\varepsilon_0}{\frac{17}{54}\frac{\rho R}{\varepsilon_0}} = \frac{9}{17}$$

Correct ans. is $\frac{18}{34}$

Q.18 Three harmonic waves having equal frequency ν and same intensity I_0, have phase angles, 0, $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ respectively. When they are superimposed the intensity of the resultant wave is close to-

(1) $3I_0$ (2) $5.8I_0$ (3) I_0 (4) $0.2I_0$

Ans. **[2]**



$I_0 \propto A_0^2$

Resultant Amplitude

$A = \sqrt{2}A_0 + A_0$

$I_0 \propto A^2$

$I_0 \propto (\sqrt{2}+1)^2 A_0^2 \Rightarrow 5.8\ I_0$

Q.19 The electric fields of two plane electromagnetic plane waves in vacuum are given by

$\vec{E}_1 = E_0\hat{j}\ \cos(\omega t - kx)$ and $\vec{E}_2 = E_0\hat{k}\cos(\omega t - ky)$

At t = 0, a particle of charge q is at origin with a velocity $\vec{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is-

(1) $E_0q(0.4\hat{i} - 3\hat{j} + 0.8\hat{k})$ (2) $E_0q(-0.8\hat{i} + \hat{j} + \hat{k})$

(3) $E_0q(0.8\hat{i} - \hat{j} + 0.4\hat{k})$ (4) $E_0q(0.8\hat{i} + \hat{j} + 0.2\hat{k})$

Ans. **[4]**

at $x = 0, t = 0$

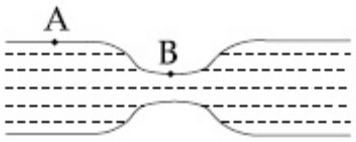
$\vec{E} = E_0\hat{j} + E_0\hat{k}$

$\vec{B} = \frac{E_0}{c}\hat{k} + \frac{E_0}{c}\hat{i}$ $\left[\text{Using } B_0 = \frac{E_0}{c},\ \hat{E}\times\hat{B} = \hat{c}\right]$

$\vec{F} = q[\vec{E} + (\vec{v}\times\vec{B})]$

$\vec{F} = qE_0[0.8\hat{i} + \hat{j} + 0.2\hat{k}]$

Q.20 Water flows in a horizontal tube (see figure). The pressure of water changes by 700 Nm^{-2} between A and B where the area of cross section are 40 cm^2 and 20 cm^2, respectively. Find the rate of flow of water through the tube. (density of water = 1000 kgm^{-3})



(Fig)

(1) 1810 cm^3/s (2) 2720 cm^3/s (3) 3020 cm^3/s (4) 2420 cm^3/s

Ans. **[2]**

$$P_A + \frac{1}{2}\rho V_A^2 = P_B + \frac{1}{2}\rho V_B^2 \qquad \{h_A = h_B\}$$

$$P_A - P_B = \frac{1}{2}\rho\left[V_B^2 - V_A^2\right] \quad ; \quad A_1V_1 = A_2V_2 \Rightarrow 2V_A = V_B$$

$$700 = \frac{1}{2}\times 10^3\left[3V_A^2\right]$$

$$V_A^2 = \frac{14}{30} \qquad \Rightarrow \qquad V_A = \sqrt{\frac{14}{30}}$$

$$\text{Flow rate} = AV_A = 40\times 10^{-4}\times\sqrt{\frac{14}{30}} \approx 2720\ cm^3/s$$

Q.21 The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x^{-n}, where n is an integer, the value of n is _____

Ans. **[3.00]**

$x^2 = at^2 + 2bt + c$ ..... (i)

$2xv = 2at + 2b$ .... (ii)

$$\Rightarrow \quad v = \frac{at + b}{x}$$

differentiating (ii) w.r.t. time

$v + x\,a' = a$

$$a' = \frac{a - v^2}{x}$$

$$a' = \frac{a - \dfrac{(at+b)^2}{x^2}}{x}$$

$$a' \propto \frac{1}{x^3}$$

$n = 3$

Q.22 A body of mass m = 10 kg is attached to one end of a wire of length 0.3 m. The maximum angular speed (in rad s^{-1}) with which it can be rotated about its other end in space station is (Breaking stress of wire = 4.8×10^7 Nm^{-2} and area of cross-section of the wire = 10^{-2} cm^2) is-

Ans. **[4.00]**

$$T = m\omega^2\ell \quad ; \quad \sigma = \frac{T}{A}$$

$$\sigma = \frac{m\omega^2\ell}{A}$$

$$\omega^2 = \frac{4.8\times10^7\times10^{-2}\times10^{-4}}{10\times0.3}$$

$\omega^2 = 16$

$\omega = 4$ rad/s

Q.23 In a fluorescent lamp choke (a small transformer) 100 V of reverse voltage is produced with the choke current changes uniformly from 0.25 A to 0 in a duration of 0.025 ms. The self-inductance of the choke (in mH) is estimated to be _____

Ans. **[10.00]**

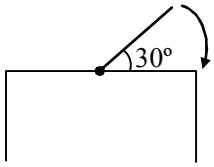
$$L\frac{di}{dt} = V$$

$$L\times\frac{0.25}{0.025\times10^{-3}} = 100$$

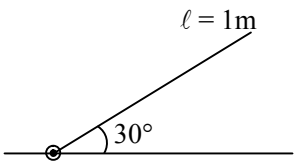
$\Rightarrow L = 10^{-2}$ H

$\Rightarrow L = 10$ mH

Q.24 One end of a straight uniform 1 m long bar is pivoted on horizontal table. It is released from rest when it makes an angle 30º from the horizontal (see figure). Its angular speed when it hits the table is given as $\sqrt{n}$ s^{-1}, where n is an integer. The value of n is _____



Ans. **[15.00]**



By energy conservation

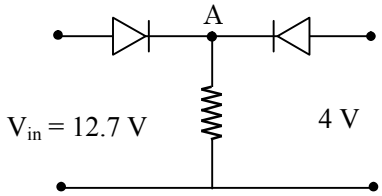
$$mgh = \frac{1}{2}I\omega^2$$

$$mg\frac{1}{2}\sin 30° = \frac{1}{2}\frac{m\ell}{3}\omega^2$$

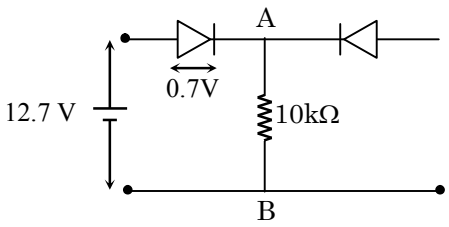
$$\omega^2 = 15 \Rightarrow \omega = \sqrt{15} \text{ rad/sec.}$$

$$n = 15$$

Q.25 Both the diodes used in the circuit shown are assumed to be ideal and have negligible resistance when these are forward biased. Built in potential in each diode is 0.7 V. For the input voltages shown in the figure, the voltage (in Volts) at point A is _____



Ans. **[12.00]**



$V_A = 12.7 - 0.7$

$V_A = 12V$

CAREER POINT

JEE Main Online Exam 2019

[Memory Based Paper]

Questions & Answer

9th January 2019 | Shift - I

CHEMISTRY

Q.1 If the magnetic moment of a dioxygen species is 1.73 B.M, it may be:

(1) O_2, O_2^- or O_2^+ (2) O_2 or O_2^- (3) $O_2 + O_2^+$ (4) O_2^- or O_2^+

Ans. **[4]**

Sol. Magnetic moment = 1.73 BM

Unpaired electron = 1

$O_2 = \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_2^2, \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^1 = \pi^* 2p_y^1$

Hence O_2^- & O_2^+ have one unpaired electron.

Q.2 The compound that cannot act both as oxidising and reducing agent is:

(1) H_2O_2 (2) H_2SO_3 (3) HNO_2 (4) H_3PO_4

Ans. **[4]**

Sol. $H_2O_2^{-1}$ $H_2SO_3^{+4}$

$H\overset{+3}{N}O_2$

$H_3\overset{+5}{P}O_4$

In H_3PO_4 phosphorous is in maximum oxidation state so can not increase its oxidation no.

Q.3 $[Pd(F)(Cl)(Br)(I)]^{2-}$ has n number of geometrical isomers. Then, the spin-only magnetic moment and crystal field stabilisation energy [CFSE] of $[Fe(CN)_6]^{n-6}$, respectively, are:

[Note: Ignore the pairing energy]

(1) 2.84 BM and $-1.6\ \Delta_0$ (2) 1.73 BM and $-2.0\ \Delta_0$

(3) 5.92 BM and 0 (4) 0 BM and $-2.4\ \Delta_0$

Ans. **[2]**

Sol. No of G.I in $[Pd\ Cl\ F\ Br\ I]^{2-}$ is 3.

Hence n = 3

So $[Fe(CN)_6]^{n-6} = [Fe(CN)_6]^{-3}$

$Fe^{3+} = 3d^5\ 4s^0$

Strong field ligand.

So,

A/C to C F T

$t_{2g}^{2,2,1}$ $eg^{0,0}$

n = 1

Magnetic moment = $\sqrt{n(n+2)} = \sqrt{1(1+2)} = \sqrt{3} = 1.73$ B.M.

CFSE = $-0.4\ \Delta_0 \times n\ t_{2g} + 0.6\ \Delta_0 \times n_{eg}$.

CFSE = $-0.4\ \Delta_0 \times 5 = -2.0\ \Delta_0$

Q.4 The electronic configurations of bivalent europium and trivalent cerium are:
(atomic number: Xe = 54, Ce = 58, Eu = 63)
(1) [Xe] $4f^7\ 6s^2$ and [Xe] $4f^2 6s^2$ (2) [Xe] $4f^7$ and [Xe] $4f^1$
(3) [Xe] $4f^2$ and [Xe] $4f^7$ (4) [Xe] $4f^4$ and [Xe] $4f^9$

Ans. **[2]**

Sol. $Eu^{2+} \rightarrow$ [xe] $4f^7$
Ce^{3+} : [xe] $4f^1$

Q.5 For following reactions

$$A \xrightarrow{700\ K} \text{Product}$$
$$A \xrightarrow[\text{catalyst}]{500\ K} \text{Product}$$

it was found that the E_a is decreased by 30 kJ/mol in the presence of catalyst. If the rate remains unchanged, the activation energy for catalysed reaction is (Assume pre exponential factor is same):
(1) 135 kJ/mol (2) 105 kJ/mol (3) 75 kJ/mol (4) 198 kJ/mol

Ans. **[2]**

Sol. $K_{catalyst} = K$

$$Ae^{-\frac{Ea_1}{RT_1}} = Ae^{-\frac{Ea_2}{RT_2}}$$

Ea_1 = energy of activation in presence of catalyst
$T_1 = 500$ K $T_2 = 700$ K

$$\frac{Ea_1}{T_1} = \frac{Ea_2}{T_2}$$

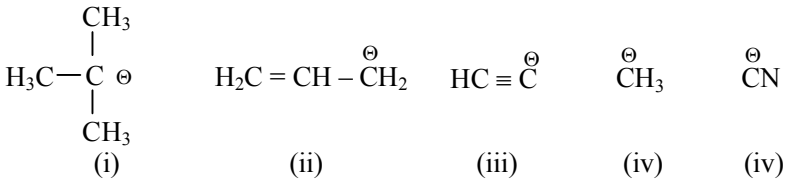
but $Ea_1 = Ea_2 - 30$

$$\frac{Ea_2 - 30}{500} = \frac{Ea_2}{700}$$

$5Ea_2 = 7Ea_2 - 210$

$$Ea_2 = \frac{210}{2} = 105 \text{ kJ/mole.}$$

Q.6 The increasing order of basicity for the following intermediates is (from weak to strong)

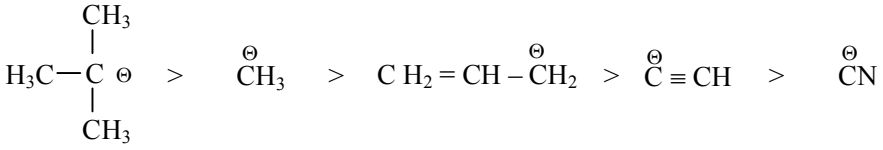


(i) (ii) (iii) (iv) (iv)

(1) (v) < (iii) < (ii) < (iv) < (i) (2) (iii) < (i) < (ii) < (iv) < (v)
(3) (v) < (i) < (iv) < (ii) < (iii) (4) (iii) < (iv) < (ii) < (i) < (v)

Ans. **[1]**

Sol. Basicity is inversly proportional to electronegativity



(i) (iv) (ii) (iii) (iv)

i > iv > ii > iii > v

Q.7 'X' melts at low temperature and is a bad conductor of electricity in both liquid and solid state. X is:
(1) Silicon carbide (2) Mercury (3) Zinc sulphide (4) Carbon tetrachloride

Ans. **[4]**

Sol. CCl_4 is a non conductor in solid and liquid phase.

Q.8 The de Broglie wavelength of an electron in the 4th Bohr orbit is:
(1) $6\pi a_0$ (2) $4\pi a_0$ (3) $2\pi a_0$ (4) $8\pi a_0$

Ans. **[4]**

Sol. We know

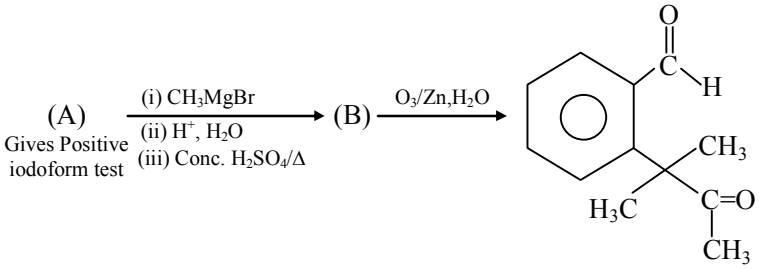
$2\pi r = n\lambda$

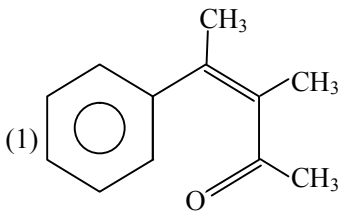
$n = 4$ & $r = a_0\frac{n^2}{z} \Rightarrow 2\pi\, a_0\frac{n^2}{z} = n\lambda$

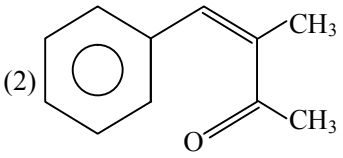
$2\pi\ \frac{4^2 a_0}{1} = 4\lambda$

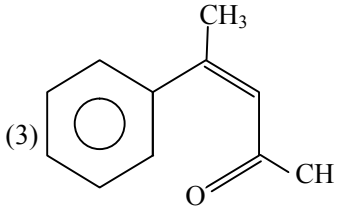
$\lambda = 8\pi a_0$

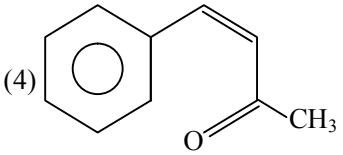
Q.9 Identify (A) in the following reaction sequence:





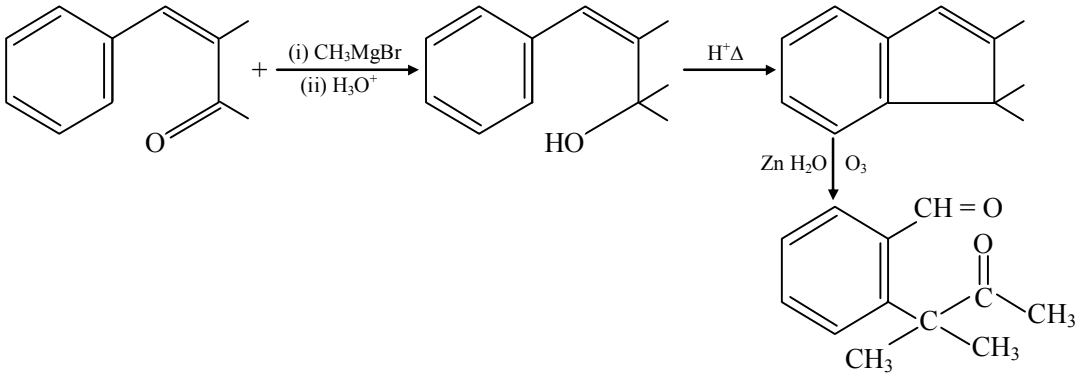




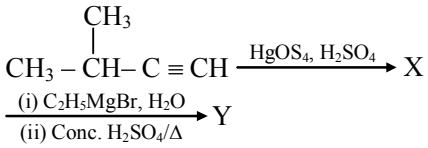


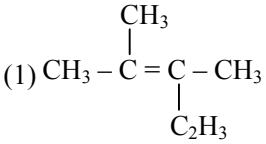
Ans. **[2]**

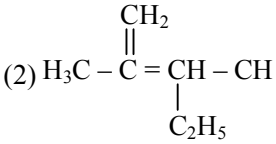
Sol.

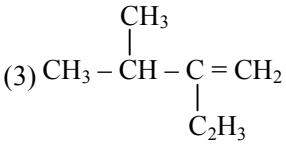


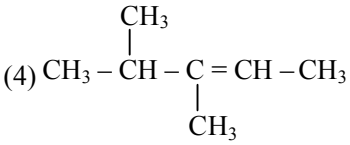
Q.10 The major product (Y) in the following reactions is:



(1) 

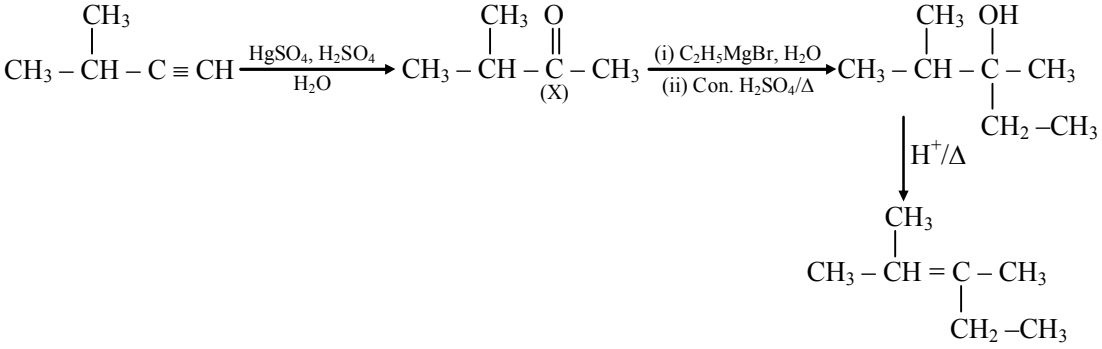
(2) 

(3) 

(4) 

Ans. **[1]**

Sol.



Q.11 Complex X of composition $Cr(H_2O)_6Cl_n$ has a spin only magnetic moment of 3.83 BM. It reacts with $AgNO_3$ and shows geometrical isomerism. The IUPAC nomenclature of X is:

(1) Dichloridotetraaqua chromium (IV) chloride dihydrate

(2) Tetraaquadichlorido chromium (III) chloride dihydrate

(3) Tetraaquadichlorido chromium (IV) chloride dihydrate

(4) Hexaaqua chromium (III) chloride

Ans. **[2]**

Sol. $Cr(H_2O)_6\ cl_n$

$(\mu_{complex})_{spin} = 3.8$ B.M.

$n = 3$

$cr(H_2O)_6\ Cl_3$

oxid^n No of Cr should be +3.

Compound So G.I so it will be

$[Cr(H_2O)_4Cl_2]\ Cl.2H_2O$

IUPAC NAME $\rightarrow$ Tetraaquadichlorido chromium(III)-chloride dihydrate

Q.12 The acidic, basic and amphoteric oxides, respectivley, are:

(1) Cl_2O, CaO, P_4O_{10} (2) MgO, Cl_2O, Al_2O_3

(3) Na_2O, SO_3, Al_2O_3 (4) N_2O_3, Li_2O, Al_2O_3

Ans. **[4]**

Sol. Non metal oxide _____ acidic

Metal oxide _____ basic

Al_2O_3 amphoteric

Q.13 The K_{sp} for the following dissociation is 1.6×10^{-5}

$$PbCl_{2(s)} \rightleftharpoons Pb^{2+}_{(aq)} + 2Cl^{-}_{(aq)}$$

Which of the following choices is correct for a mixture of 300 mL 0.134 M $Pb(NO_3)_2$ and 100 mL 0.4 M NaCl?

(1) Not enough data provided
(2) $Q > K_{sp}$
(3) $Q < K_{sp}$
(4) $Q = K_{sp}$

Ans. **[2]**

Sol. $Q = [Pb^{2+}][Cl^{-}]^2$

$$= \left(\frac{300 \times 0.134}{400}\right)\left(\frac{100 \times 0.4}{400}\right)^2$$

$$= \frac{3 \times 0.134}{4}(0.1)^2$$

$Q = 1.005 \times 10^{-3}$

$Q > k_{sp}$.

Q.14 A chemist has 4 samples of artificial sweetener A, B, C and D. To identify thèse samples, he performed certain experiments and noted the following observations:

(i) A and D both form blue-violet colour with ninhydrin.
(ii) Lassaigne extract of C gives positive $AgNO_3$ test and negative $Fe_4[Fe(CN)_6]_3$ test.
(iii) Lassaigne extract of B and D gives positive sodium nitroprusside test.

Based on thèse observations which option is correct?

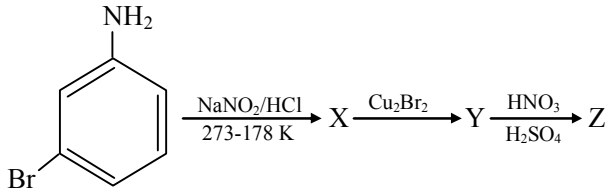
(1) A : Aspartame; B : Alitame; C : Saccharin; D : Sucralose
(2) A : Aspartame; B : Saccharin; C : Sucralose; D : Alitame
(3) A : Alitame; B : Saccharin; C : Aspartame; D : Sucralose
(4) A : Saccharin; B : Alitame; C : Sucralose; D : Aspartame

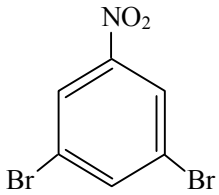
Ans. **[2]**

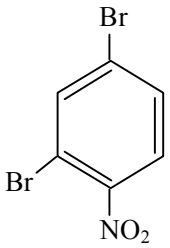
Sol. A – Aspartame
B – Saccharine
C – Sucralose
D – Alitame

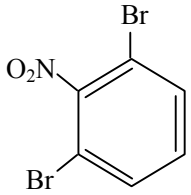
(i) A & D give positive test with ninhydrin because both have free carboxylic and amine group.
(ii) C form precipitate with $AgNO_3$ because it has chlorine atoms
(iii) B & D give positive test because Both have S–atom.

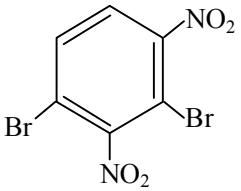
Q.15 The major product Z obtained in the following reaction scheme is:



(1) 

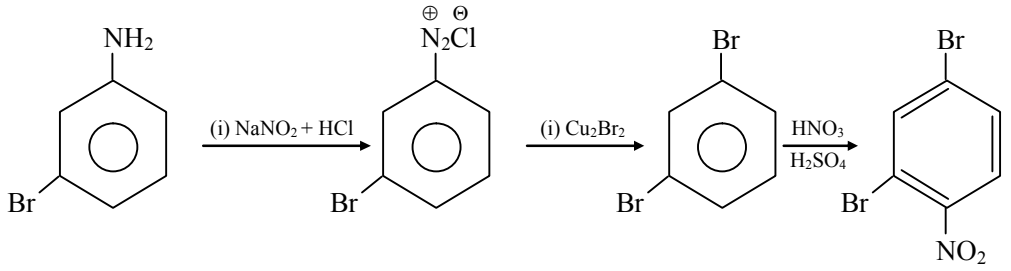
(2) 

(3) 

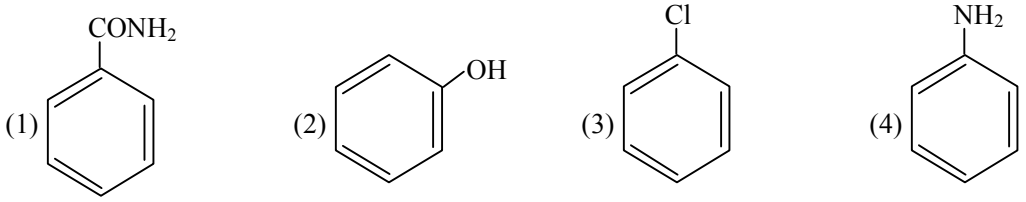
(4) 

Ans. **[2]**

Sol.



Q.16 Which of thèse will produce the highest yield in Friedel Crafts reaction?



Ans. **[2]**

Sol. Aniline react lewis acid form anilinium complex.

So phenol is most reactive among for nucleophellic substitution reaction.

Q.17 B has a smaller first ionization enthalpy than Be. Consider the following statements:

(i) it is easier to remove 2p electron than 2s electron

(ii) 2p electron of B is more shielded from the nucleus by the inner core of electrons than the 2s electrons of Be

(iii) 2s electron has more penetration power than 2p electron

(iv) atomic radius of B is more than Be

(atomic number B = 5, Be = 4)

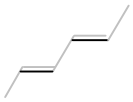
The correct statements are:

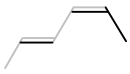
(1) (i), (iii) and (iv) (2) (i), (ii) and (iii) (3) (i), (ii) and (iv) (4) (ii), (iii) and (iv)

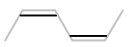
Ans. **[4]**

Sol. Radius of Boron is less than Radius of Be Hence IV statement is false. Rest are True. Statement

Q.18 The correct order of heat of combustion for following alkadienes is:

(i) 

(ii) 

(iii) 

(1) (i) < (ii) < (iii) (2) (i) < (iii) < (ii) (3) (iii) < (ii) < (i) (4) (ii) < (iii) < (i)

Ans. **[1]**

Sol. In lsomers of Hydrocarbon heat of combustion depend upon their stability. Stability increase heat of combustion decrease.

Stability a > b > c

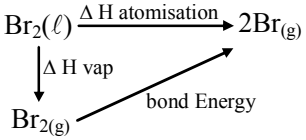
Heat of combustion c > b > a

Q.19 If enthalpy of atomisation for $Br_{2(l)}$ is x kJ/mol and bond enthalpy for Br_2 is y kJ/mol, the relation between them:

(1) is x = 7y (2) is x > y (3) does not exist (4) is x < y

Ans. **[2]**

Sol.

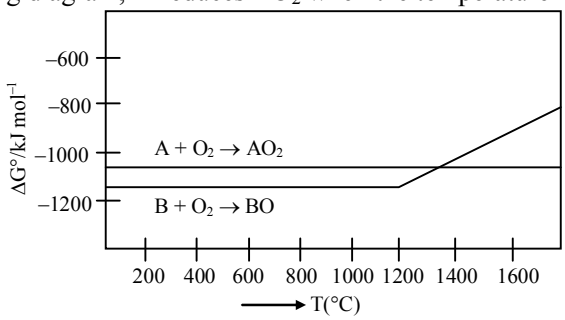


Δ H atomisation = Δ H Vap + Bond Energy

Hence.

x > y

Q.20 According to the following diagram, A reduces BO_2 when the temperature is:



(1) > 1400 °C (2) < 1400 °C (3) < 1200 °C (4) > 1200 °C but < 1400 °C

Ans. **[1]**

Sol. $A + BO_2 \rightarrow B + AO_2$

ΔG = –ve

Only above 1400 °C.

Q.21 The molarity of HNO in a sample which has density 1.4 g/mL and mass percentage of 63% is _____. (Molecular Weight of HNO_3 = 63)

Ans. **[14.00]**

Sol. d = 1.4 g/ml

63/w/w

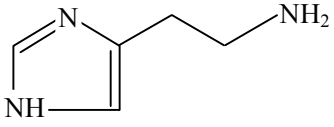
$$\text{Molarity} = \frac{63\times1.4}{63\times100} \times 1000 \text{ mole/litre}$$

= 14 mole/L.

Q.22 The mass percentage of nitrogen In histamine is _____.

Ans. **[37.84]**

Sol. Structure of Histamine is



Molecular formula = $C_5H_9N_3$

Molecular mass = 111

$$\% \text{ of N} = \frac{42}{111} \times 100$$

= 37.84 %

Q.23 How much amount of NaCl should be added to 600 g of watre (ρ = 1.00 g/mL) to decrease the freezing point of water to –0.2°C ? _____. (The freezing point depression constant for water = 2 K kg mol^{-1})

Ans. **[01.76]**

Sol. $\Delta T_f = 0.2$

$\Delta T_f = i\, K_f\, m$

i = 2 for NaCl

$0.2 = 2 \times 2 \times m$

$m = \frac{w}{58.5} \times \frac{1000}{600}$

$0.2 = 2 \times 2 \times \frac{w}{58.5} \times \frac{1000}{600}$

$w = \frac{0.2 \times 58.5 \times 600}{4 \times 1000} = \frac{1.2 \times 58.5}{40} = 01.76$ g

Q.24 108 g of silver (molar mass 108 g mol^{-1}) is deposited at cathode from $AgNO_3$(aq) solution by a certain quantity of electricity. The volume (in L) of oxygen gas produced at 273 K and 1 bar pressure from water by the same quantity of electricity is_____.

Ans. **[0.568]**

Sol. $[n_{Ag}]_{deposit} = \frac{108}{108}$

= 1 mole.

$Ag^+ + 1e \rightarrow Ag$

1F charge is required for 1 mole of Ag

$H_2O \rightarrow \frac{1}{2} O_2 + 2H^+ + 2e^-$

2F charge deposit _____ $\frac{1}{2}$ mole

1F charge deposit _____ $\frac{1}{4}$ mole

PV = n R T

$V = \frac{1}{4} \times \frac{0.83 \times 273}{1}$

V = 5.675 L

Q.25 The hardness of a water sample containing 10^{-3} M $MgSO_4$ expressed as $CaCO_3$ equivalents (in ppm) is _____.

(molar mass of $MgSO_4$ is 120.38 g/mol)

Ans. **[100.00]**

Sol. Given: 10^{-3} M $MgSO_4$

i.e. 10^{-3} mole $MgSO_4$ present in 1 litre solution

$n_{MgSO_4} = n_{ca\,co_3}$

Mass of $CaCO_3 = 10^{-3} \times 100$

ppm (in terms of $CaCO_3$) = $\frac{10^{-3} \times 100}{1000} \times 10^6$

= 100 ppm

CAREER POINT

JEE Main Online Exam 2020

Questions & Solution

9th January 2020 | Shift - I

MATHEMATICS

Q.1 In a box, there are 20 cards, out of which 10 are labelled as A and the remaining 10 are labelled as B. Cards are drawn at random, one after the other and with replacement, till a second A-card is obtained. The probability that the second A-card appears before the third B-card is

(1) $\frac{11}{16}$ (2) $\frac{15}{16}$ (3) $\frac{9}{16}$ (4) $\frac{13}{16}$

Ans. **[1]**

Sol. Possibilities that the second A card appears before the third B card are

$= AA + ABA + BAA + ABBA + BBAA + BABA$

$= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4 + \left(\frac{1}{2}\right)^4$

$= \frac{1}{4} + \frac{1}{8} + \frac{1}{8} + \frac{1}{16} + \frac{1}{16} + \frac{1}{16}$

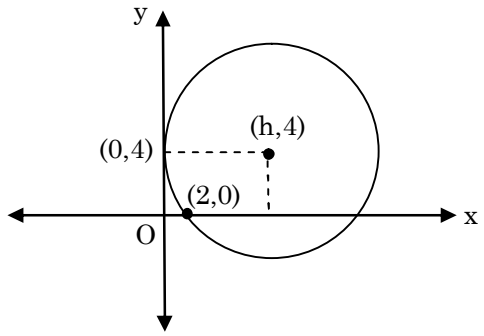
$= \frac{11}{16}$

Q.2 A circle touches the y-axis at the point (0, 4) and passes through the point (2, 0). Which of the following lines is not a tangent to this circle?

(1) $3x - 4y - 24 = 0$ (2) $4x - 3y + 17 = 0$ (3) $4x + 3y - 8 = 0$ (4) $3x + 4y - 6 = 0$

Ans. **[4]**

Sol. Let the centre of circle is (h, 4) and r = h



$\therefore$ equation of circle $(x - h)^2 + (y - 4)^2 = h^2$

it passes through (2, 0)

$\therefore (2 - h)^2 + 16 = h^2$

$\Rightarrow h = 5$

$\therefore$ centre (5, 4) and r = 5.

By option $3x + 4y - 6 = 0$ is a tangent to the circle ($\because$ p = r)

$$p = \left|\frac{15+16-6}{\sqrt{9+16}}\right| = 5$$

Q.3 If for some α and β in R, the intersection of the following three planes
$x + 4y - 2z = 1$
$x + 7y - 5z = \beta$
$x + 5y + \alpha z = 5$
is a line in R^3, then $\alpha + \beta$ is equal to
(1) 0 (2) –10 (3) 10 (4) 2

Ans. **[3]**

Sol. Plane intersect in a line

$$\therefore \Delta = \begin{vmatrix} 1 & 4 & -2 \\ 1 & 7 & -5 \\ 1 & 5 & \alpha \end{vmatrix} = 0$$

$\Rightarrow 1(7\alpha + 25) - 4(\alpha + 5) - 2(5 - 7) = 0$

$\Rightarrow 7\alpha + 25 - 4\alpha - 20 + 4 = 0$

$\Rightarrow 3\alpha + 9 = 0$

$\Rightarrow \alpha = -3$

Also $\Delta_z = 0$

$$\Rightarrow \begin{vmatrix} 1 & 4 & 1 \\ 1 & 7 & \beta \\ 1 & 5 & 5 \end{vmatrix} = 0$$

$\Rightarrow 1(35 - 5\beta) - 4(5 - \beta) + 1(5 - 7) = 0$

$\Rightarrow 35 - 5\beta - 20 + 4\beta - 2 = 0$

$\Rightarrow \beta = 13$

$\therefore \alpha + \beta = 10$

Q.4 A spherical iron ball of 10 cm radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3/min. When the thickness of ice is 5 cm, then the rate (in cm/min.) at which of the thickness of ice decreases, is

(1) $\frac{1}{18\pi}$ (2) $\frac{1}{54\pi}$ (3) $\frac{1}{36\pi}$ (4) $\frac{5}{6\pi}$

Ans. **[1]**

Sol. Let the thickness = x cm

Total volume $V = \frac{4}{3}\pi(10 + x)^3$

$$\frac{dV}{dt} = \frac{4}{3}\pi\, 3(10+x)^2 \frac{dx}{dt}$$

Given that $\frac{dV}{dt} = 50\ cm^3/min,\ x = 5\ cm$

$$\Rightarrow 50 = \frac{4}{3}\pi\, 3(10+5)^2 \frac{dx}{dt}$$

$$\Rightarrow \frac{dx}{dt} = \frac{1}{18\pi}\ cm/min.$$

Q.5 If the matrices $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{bmatrix}$, B = adj A and C = 3A, then $\frac{|adj\ B|}{|C|}$ is equal to :

(1) 16 (2) 8 (3) 2 (4) 72

Ans. **[2]**

Sol. $|A| = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 3 & 4 \\ 1 & -1 & 3 \end{vmatrix} = 6$

$|adj\ B| = |adj\ adj\ A| = |A|^{(n-1)^2} = |A|^4 = 36 \times 36$

$|C| = 27|A| = 27 \times 6$

$$\frac{|adj\ B|}{|C|} = \frac{36\times 36}{27\times 6} = 8$$

Q.6 If $f(x) = \begin{cases} \frac{\sin(a+2)x + \sin x}{x} & ; \quad x < 0 \\ b & ; \quad x = 0 \\ \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}} & ; \quad x > 0 \end{cases}$ is continuous at x = 0, then a + 2b is equal to :

(1) –2 (2) –1 (3) 1 (4) 0

Ans. **[4]**

Sol. $L.H.L. = \lim\limits_{x\to 0} \frac{\sin(a+2)x + \sin x}{x}$

$= a + 3$

$$R.H.L. = \lim_{x\to 0} \frac{(x+3x^2)^{1/3} - x^{1/3}}{x^{4/3}} = \lim_{x\to 0} \frac{(1+3x)^{1/3} - 1}{x}$$

$$= \lim_{x\to 0} \frac{\frac{1}{3}(1+3x)^{-2/3}.3}{1} = 1$$

$f(0) = b$

$b = 1$ and $a = -2$, So $a + 2b = 0$

Q.7 The value of $\int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$ is equal to

(1) $2\pi^2$ (2) 4π (3) π^2 (4) 2π

Ans. **[3]**

Sol. $I = \int_0^{2\pi} \frac{x\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$ ....(1)

$I = \int_0^{2\pi} \frac{(2\pi-x)\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$ ....(2)

Eq. (1) + (2)

$2I = \int_0^{2\pi} \frac{2\pi\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$

$I = 2\pi\int_0^{\pi} \frac{\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$

$I = 4\pi\int_0^{\pi/2} \frac{\sin^8 x}{\sin^8 x+\cos^8 x}\,dx$ ...(3)

$I = 4\pi\int_0^{\pi/2} \frac{\cos^8 x}{\cos^8 x+\sin^8 x}\,dx$ ...(4)

Eq. (3) + (4)

$2I = 4\pi\int_0^{\pi/2} dx = \pi^2$

Q.8 The integral $\int \frac{dx}{(x+4)^{8/7}(x-3)^{6/7}}$ is equal to : (where C is a constant of integration)

(1) $-\frac{1}{13}\left(\frac{x-3}{x+4}\right)^{-13/7}+C$ (2) $-\left(\frac{x-3}{x+4}\right)^{-1/7}+C$ (3) $\frac{1}{2}\left(\frac{x-3}{x+4}\right)^{3/7}+C$ (4) $\left(\frac{x-3}{x+4}\right)^{1/7}+C$

Ans. **[4]**

Sol. $\int \frac{1}{(x+4)^{8/7}(x-3)^{6/7}}\,dx$

$= \int \frac{dx}{(x+4)^2\left(\frac{x-3}{x+4}\right)^{6/7}}$

Put $\frac{x-3}{x+4} = t$

$$\left\{\frac{(x+4)-(x-3)}{(x+4)^2}\right\}dx = dt$$

$$= \frac{1}{7}\int \frac{dt}{t^{6/7}}$$

$$= \frac{1}{7}\left(\frac{t^{1/7}}{1/7}\right)+C$$

$$= \left(\frac{x-3}{x+4}\right)^{1/7}+C$$

Q.9 Let f be any function continuous on [a, b] and twice differentiable on (a, b). If for all $x \in (a, b)$, $f'(x) > 0$ and $f''(x) < 0$, then for any $c \in (a, b)$, $\frac{f(c)-f(a)}{f(b)-f(c)}$ is greater than :

(1) $\frac{b+a}{b-a}$ (2) 1 (3) $\frac{c-a}{b-c}$ (4) $\frac{b-c}{c-a}$

Ans. **[3]**

Sol. Use LMVT for $x \in [a, c]$

$$\frac{f(c)-f(a)}{c-a} = f'(\alpha),\ \alpha \in (a, c)$$

also use LMVT for $x \in [c, b]$

$$\frac{f(b)-f(c)}{b-c} = f'(\beta);\ \beta \in (c, b)$$

given : $f''(x) < 0$

$\Rightarrow f'(x)$ is decreasing $f'(\alpha) < f'(\beta)$

$$\frac{f(c)-f(a)}{c-a} > \frac{f(b)-f(c)}{b-c}$$

$$\frac{f(c)-f(a)}{f(b)-f(c)} > \frac{c-a}{b-c} \qquad (\because f(x) \text{ is increasing})$$

Q.10 The number of real roots of the equation $e^{4x} + e^{3x} - 4e^{2x} + e^{x} + 1 = 0$ is

(1) 4 (2) 1 (3) 3 (4) 2

Ans. **[2]**

Sol. Let $e^x = t \in (0, \infty)$

given eq. $\Rightarrow t^4 + t^3 - 4t^2 + t + 1 = 0$

$$t^2 + t - 4 + \frac{1}{t} + \frac{1}{t^2} = 0$$

$$\left(t^2 + \frac{1}{t^2}\right)+\left(t + \frac{1}{t}\right) - 4 = 0$$

Let $t + \frac{1}{t} = z$

$(z^2 - 2) + (z) - 4 = 0$

$z^2 + z - 6 = 0$

$z = -3, 2$

$\Rightarrow z = 2 \Rightarrow e^x + e^{-x} = 2$

$x = 0$ is only solution.

Q.11 Let z be a complex number such that $\left|\frac{z-i}{z+2i}\right| = 1$ and $|z| = \frac{5}{2}$. Then the value of $|z + 3i|$ is :

(1) $\sqrt{10}$ (2) $2\sqrt{3}$ (3) $\frac{15}{4}$ (4) $\frac{7}{2}$

Ans. **[4]**

Sol. $|z - i| = |z + 2i|$ (let $z = x + iy$)

$\Rightarrow x^2 + (y - 1)^2 = x^2 + (y + 2)^2$

$\Rightarrow y = -\frac{1}{2}$

again $|z| = \frac{5}{2}$

$x^2 + y^2 = \frac{25}{4}$

$\Rightarrow x^2 = 6$

$z = \pm\sqrt{6} - \frac{1}{2}i$

$|z + 3i| = \sqrt{6 + \frac{25}{4}} = \frac{7}{2}$

Q.12 The value of $\cos^3\left(\frac{\pi}{8}\right).\cos\left(\frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right).\sin\left(\frac{3\pi}{8}\right)$ is

(1) $\frac{1}{4}$ (2) $\frac{1}{2\sqrt{2}}$ (3) $\frac{1}{2}$ (4) $\frac{1}{\sqrt{2}}$

Ans. **[2]**

Sol. $\cos^3\left(\frac{\pi}{8}\right).\sin\left(\frac{\pi}{2} - \frac{3\pi}{8}\right) + \sin^3\left(\frac{\pi}{8}\right).\cos\left(\frac{\pi}{2} - \frac{3\pi}{8}\right)$

$= \cos^3\frac{\pi}{8}.\sin\frac{\pi}{8} + \sin^3\frac{\pi}{8}.\cos\frac{\pi}{8}$

$= \cos\frac{\pi}{8}.\sin\frac{\pi}{8}\left(\cos^2\frac{\pi}{8}.\sin^2\frac{\pi}{8}\right)$

$= \frac{1}{2}\left(2\cos\frac{\pi}{8}.\sin\frac{\pi}{8}\right)$

$= \frac{1}{2}\sin\frac{\pi}{4}$

$= \frac{1}{2\sqrt{2}}$

Q.13 If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{18}+\frac{y^2}{4} = 1$ and the hyperbola, $\frac{x^2}{9}-\frac{y^2}{4} = 1$ respectively and (e_1, e_2) is a point on the ellipse, $15x^2 + 3y^2 = k$, then k is equal to

(1) 16 (2) 14 (3) 15 (4) 17

Ans. **[1]**

Sol. $e_1 = \sqrt{1-\frac{4}{18}} = \frac{\sqrt{7}}{3}$

$e_2 = \sqrt{1+\frac{4}{9}} = \frac{\sqrt{13}}{3}$

(e_1, e_2) lies on $15x^2 + 3y^2 = k$

$\therefore\ 15\left(\frac{7}{9}\right) + 3\left(\frac{13}{9}\right) = k$

$\Rightarrow\ k = 16$

Q.14 If $f'(x) = \tan^{-1}(\sec x + \tan x), -\frac{\pi}{2} < x < \frac{\pi}{2}$, and $f(0) = 0$, then $f(1)$ is equal to

(1) $\frac{\pi-1}{4}$ (2) $\frac{1}{4}$ (3) $\frac{\pi+2}{4}$ (4) $\frac{\pi+1}{4}$

Ans. **[4]**

Sol. $f'(x) = \tan^{-1}(\sec x + \tan x);\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f'(x) = \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right) = \tan^{-1}\left\{\frac{1-\cos\left(\frac{\pi}{2}+x\right)}{\sin\left(\frac{\pi}{2}+x\right)}\right\}$

$f'(x) = \tan^{-1}\left\{\frac{2\sin^2\left(\frac{\pi}{4}+\frac{x}{2}\right)}{2\sin\left(\frac{\pi}{4}+\frac{x}{2}\right)\cos\left(\frac{\pi}{4}+\frac{x}{2}\right)}\right\}$

$f'(x) = \tan^{-1}\left\{\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right\} = \frac{\pi}{4}+\frac{x}{2}$

integrating on both side

$f(x) = \frac{\pi}{4}x + \frac{x^2}{4} + c$

given that $f(0) = 0$

$\Rightarrow c = 0$

So, $f(x) = \frac{\pi}{4}x + \frac{x^2}{4}$

$f(1) = \frac{\pi}{4} + \frac{1}{4} = \frac{\pi+1}{4}$

Q.15 Let the observations $x_i (1 \le i \le 10)$ satisfy the equations, $\sum_{i=1}^{10}(x_i - 5) = 10$ and $\sum_{i=1}^{10}(x_i - 5)^2 = 40$. If μ and λ are the mean and the variance of the observations, $x_1 - 3$, $x_2 - 3$, ... $x_{10} - 3$, then the ordered pair (μ, λ) is equal to

(1) (3, 6) (2) (6, 3) (3) (3, 3) (4) (6, 6)

Ans. **[3]**

Sol. Mean of $(x_i - 5) = \frac{\Sigma(x_i - 5)}{10} = 1$

Now, mean of $(x_i - 3)$ = mean of $\{(x_i - 5) + 2\} = 1 + 2$

$\lambda = 3$

μ = variance of $(x_i - 5) = \frac{\Sigma(x_i - 5)^2}{10} - (\text{mean})^2$

$= \frac{40}{10} - (1)^2 = 3$

Q.16 Negation of the statement : '$\sqrt{5}$ is an integer or 5 is irrational ' is

(1) $\sqrt{5}$ is not an integer or 5 is not irrational (2) $\sqrt{5}$ is irrational or 5 is an integer

(3) $\sqrt{5}$ is an integer and 5 is irrational (4) $\sqrt{5}$ is not an integer and 5 is not irrational

Ans. **[4]**

Sol. $\sqrt{5}$ is not an integer and 5 is not irrational

Q.17 If the number of five digit numbers with distinct digits and 2 at the 10th place is 336 k, then k is equal to

(1) 7 (2) 4 (3) 6 (4) 8

Ans. **[4]**

Sol.

			2	

$8 \times 8 \times 7 \times 6 = 2688 = 336k$

$\Rightarrow k = 8$

Q.18 If for all real triplets (a, b, c), $f(x) = a + bx + cx^2$; then $\int_0^1 f(x)\,dx$ is equal to :

(1) $\frac{1}{2}\left\{f(1)+3f\left(\frac{1}{2}\right)\right\}$ (2) $\frac{1}{3}\left\{f(0)+f\left(\frac{1}{2}\right)\right\}$

(3) $2\left\{3f(1)+2f\left(\frac{1}{2}\right)\right\}$ (4) $\frac{1}{6}\left\{f(0)+f(1)+4f\left(\frac{1}{2}\right)\right\}$

Ans. **[4]**

Sol. $\int_0^1 (a+bx+cx^2)\,dx = \left(ax+\frac{bx^2}{2}+\frac{cx^3}{3}\right)_0^1 = a+\frac{b}{2}+\frac{c}{3}$

$f(1) = a + b + c$

$f(0) = a$

$f\left(\frac{1}{2}\right) = a+\frac{b}{2}+\frac{c}{4}$

Now, $\frac{1}{6}\left[f(0)+f(1)+4f\left(\frac{1}{2}\right)\right] = \frac{1}{6}(6a+3b+2c) = a+\frac{b}{2}+\frac{c}{3}$

Q.19 Let C be the centroid of the triangle with vertices (3, –1), (1, 3) and (2, 4). Let P be the point of intersection of the lines $x + 3y – 1 = 0$ and $3x – y + 1 = 0$. Then the line passing through the points C and P also passes through the point :

(1) (–9, –6) (2) (9, 7) (3) (–9, –7) (4) (7, 6)

Ans. **[1]**

Sol. Centroid D $\left(\frac{3+1+2}{3}, \frac{-1+3+4}{3}\right) \equiv (2, 2)$

Point of intersection of lines $x + 3y – 1 = 0$ and $3x – y + 1 = 0$ is P $\left(-\frac{1}{5}, \frac{2}{5}\right)$

So, equation of line DP is $8x – 11y + 6 = 0$

Point (–9, –6) satisfy this equation.

Q.20 The product $2^{\frac{1}{4}}\cdot 4^{\frac{1}{16}}\cdot 8^{\frac{1}{48}}\cdot 16^{\frac{1}{128}}\ldots.$ to ∞ is equal to :

(1) $2^{1/2}$ (2) 2 (3) 1 (4) $2^{1/4}$

Ans. **[1]**

Sol. $2^{\frac{1}{4}}\cdot 4^{\frac{1}{16}}\cdot 8^{\frac{1}{48}}\cdot 16^{\frac{1}{128}}\ldots.$

$= 2^{\frac{1}{4}+\frac{2}{16}+\frac{3}{48}+\ldots\ldots\infty}$

$= 2^{\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\ldots\ldots\infty}$

$= 2^{\left(\frac{1/4}{1-(1/2)}\right)} = 2^{1/2}$

Q.21 The projection of the line segment joining the points (1, –1, 3) and (2, –4, 11) on the line joining the points (–1, 2, 3) and (3, –2, 10) is _____

Ans. **[8]**

Sol. Let A (1, – 1, 3), B(2, – 4, 11), C (–1, 2, 3) & D (3, –2, 10)

$\overrightarrow{AB} = \hat{i} - 3\hat{j} + 8\hat{k}$

$\overrightarrow{CD} = 4\hat{i} - 4\hat{j} + 7\hat{k}$

Projection of $\overrightarrow{AB}$ on $\overrightarrow{CD}$ $= \dfrac{\overrightarrow{AB}.\overrightarrow{CD}}{|\overrightarrow{CD}|} = \dfrac{4+12+56}{\sqrt{16+16+49}} = \dfrac{72}{9} = 8$

Q.22 If for $x \geq 0$, $y = y(x)$ is the solution of the differential equation, $(x + 1)dy = ((x + 1)^2 + y - 3)\,dx$, $y(2) = 0$, then $y(3)$ is equal to _____

Ans. **[3]**

Sol. $(1 + x)\dfrac{dy}{dx} - y = (1 + x)^2 - 3$

$\Rightarrow \dfrac{dy}{dx} - \dfrac{y}{1+x} = (1 + x) - \dfrac{3}{1+x}$

I.F. $= -e^{-\int \frac{1}{1+x}dx} = \dfrac{1}{1+x}$

$\therefore$ solution is $y\left(\dfrac{1}{1+x}\right) = \int\left((1+x) - \dfrac{3}{1+x}\right)\dfrac{1}{1+x}\,dx$

$\dfrac{y}{1+x} = x + \dfrac{3}{1+x} + c$

Put $x = 2$, $y = 0 \Rightarrow c = -3$

$\Rightarrow \dfrac{y}{1+x} = x + \dfrac{3}{1+x} - 3$

Put $x = 3 \Rightarrow \dfrac{y}{4} = 3 + \dfrac{3}{4} - 3 \Rightarrow y = 3$

Q.23 The coefficient of x^4 in the expansion of $(1 + x + x^2)^{10}$ is _____

Ans. **[615]**

Sol. General term $= \dfrac{10!}{\alpha!\,\beta!\,\gamma!}x^{\beta+2\gamma}$

For coeff. of $x^4 \Rightarrow \beta + 2\gamma = 4 \qquad (\alpha + \beta + \gamma = n)$

$\gamma = 0, \beta = 4, \alpha = 6 \Rightarrow \dfrac{10!}{6!\,4!\,0!} = 210$

$\gamma = 1, \beta = 2, \alpha = 7 \Rightarrow \dfrac{10!}{7!\,2!\,1!} = 360$

$\gamma = 2, \beta = 0, \alpha = 8 \Rightarrow \dfrac{10!}{8!\,0!\,2!} = 45$

Total = 615

Q.24 The number of distinct solutions of the equation, $\log_{1/2}|\sin x| = 2 - \log_{1/2}|\cos x|$ in the interval $[0, 2\pi]$, is _____

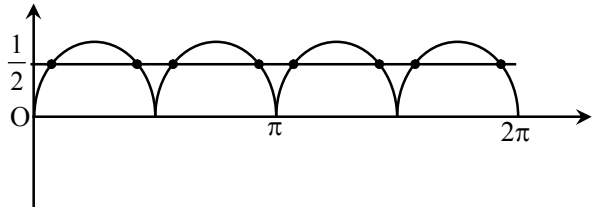
Ans. **[8]**

Sol. $\log_{1/2}|\sin x| + \log_{1/2}|\cos x| = 2$

$\Rightarrow \log_{1/2}|\sin x||\cos x| = 2$

$\Rightarrow |\sin x||\cos x| = \left(\frac{1}{2}\right)^2$

$\Rightarrow |\sin 2x| = \frac{1}{2}$



8 solutions

Q.25 If the vectors, $\vec{p} = (a+1)\hat{i} + a\hat{j} + a\hat{k}$, $\vec{q} = a\hat{i} + (a+1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a+1)\hat{k}$ $(a \in R)$ are coplanar and $3(\vec{p}.\vec{q})^2 - \lambda\left|\vec{r}\times\vec{q}\right|^2 = 0$, then the value of λ is _____

Ans. **[1]**

Sol. $\begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$

$\Rightarrow (a+1) + a + a = 0$

$\Rightarrow a = -\frac{1}{3}$

$\vec{p} = \frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$

$\vec{q} = -\frac{1}{3}\hat{i} + \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$

$\vec{r} = -\frac{1}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k}$

$\vec{p}\cdot\vec{q} = \frac{1}{9}(-2-2+1) = -\frac{1}{3}$

$\vec{r}\times\vec{q} = \frac{1}{9}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{vmatrix}$

$= \frac{1}{9}(3\hat{i} + 3\hat{j} + 3\hat{k}) = \frac{\hat{i}+\hat{j}+\hat{k}}{3}$

$|\vec{r}\times\vec{q}|^2 = \frac{1}{3}$

$\Rightarrow 3(\vec{p}\cdot\vec{q})^2 - \lambda|\vec{r}\times\vec{q}|^2 = 0$

$\Rightarrow 3\left(\frac{1}{9}\right) - \lambda\left(\frac{1}{3}\right) = 0$

$\Rightarrow \lambda = 1$